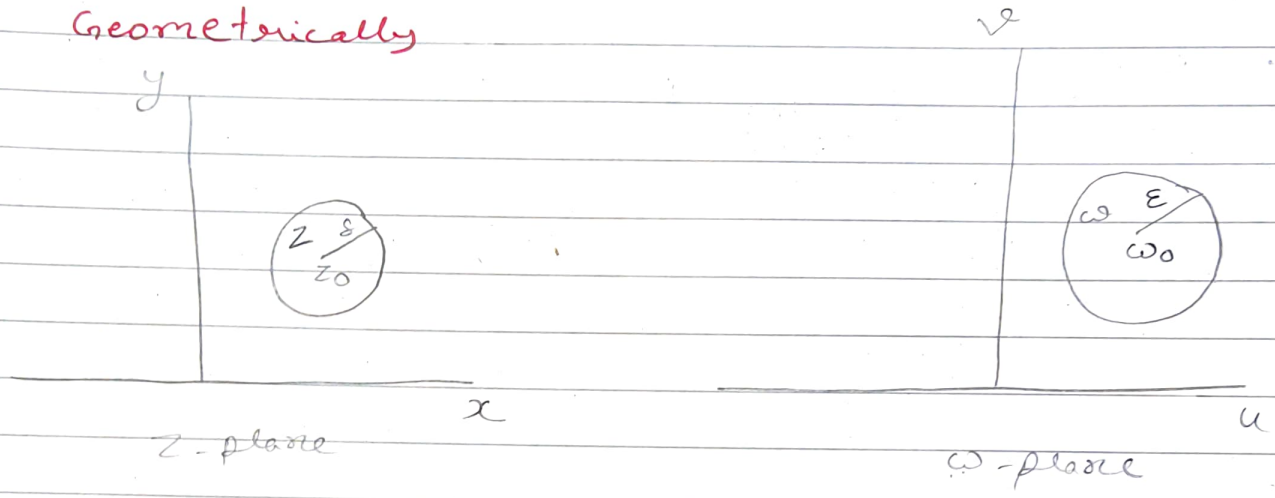
Limit

Limits let a function be defined at all points z in some deleted nbd of z_0.
w_0 is said to be the limit of function $f(z)$ at z_0 if for each $\varepsilon > 0 \exists$ a $\delta > 0$ s.t.

$$|f(z) - w_0| < \varepsilon \quad \text{whenever } 0 < |z - z_0| < \delta.$$

Geometrically



limit of a function is unique, if it exists.

Example 1. $f(z) = iz/2$ in open disk $|z| < 1$ then show that $\lim_{z \to 1} f(z) = \frac{i}{2}$

point 1 is on the boundary of the domain in disk $|z| < 1$

$$\left| f(z) - \frac{i}{2} \right| = \left| \frac{i\bar{z}}{2} - \frac{i}{2} \right| = \left| \frac{i(x - iy) - i}{2} \right|$$

$$= \left| \frac{y + i(x-1)}{2} \right| = \frac{1}{2} |y + i(x-1)| = \frac{1}{2} |(x-1) + iy|$$

$$= \frac{1}{2} |z - 1|$$

Let $\delta = 2\varepsilon$

then $\left| f(z) - \frac{i}{2} \right| < \varepsilon$ whenever $0 < |z - 1| < 2\varepsilon$

Example 2. If $f(z) = \frac{z}{\bar{z}}$ the $\lim_{z \to 0} f(z)$ does not exist.

If it exists, it could be found by letting the point $z(x,y)$ approach the origin in any manner.

Let we go through real axis, $z = (x,0)$ is a non-zero point

$$f(z) = \frac{x+iy}{x-iy} = \frac{x+i0}{x-i0} = 1$$

If we go through the imaginary axis, take $z(0,y)$ a non-zero point

$$f(z) = \frac{0+iy}{0-iy} = -1$$

Since limit is unique, limit doesn't exist for this $f(z)$.

Continuity

Continuity A function f is continuous at a point z_0 if

$\lim_{z \to z_0} f(z)$ exists ; $f(z_0)$ exists and

$$\lim_{z \to z_0} f(z) = f(z_0)$$

or

$f(z)$ is said to be continuous at z_0 if for each $\varepsilon > 0$, $\exists\, \delta > 0$ s.t.

$$|f(z) - f(z_0)| < \varepsilon \quad \text{whenever} \quad |z - z_0| < \delta$$

A function is said to **continuous in a region R** if it is continuous at each point of R

TS4

Q/2 Check the continuity for function at $z=0$.

(i) $f(0) = 0$ and

$$f(z) = \frac{Re(z^2)}{|z|} ; \quad z \neq 0$$

$$= \frac{x^2 - y^2}{\sqrt{x^2+y^2}}$$

Let $x = r\cos\theta$; $y = r\sin\theta$; $r \to 0$

$$f(z) = r(\cos^2\theta - \sin^2\theta)$$

$$= r\cos 2\theta$$

If $z \to 0$; $r \to 0$ $\lim_{r \to 0} f(z) = 0$. [limit exists and 0].

$= f(0)$.

function is continuous.

Ex. $|z|^2 \operatorname{Img}\left(\frac{1}{z}\right) = (x^2+y^2)\left(\frac{-y}{x^2+y^2}\right) = -y$

at $z=0$

as $(x,y) \to (0,0)$ $\quad \lim_{z\to 0} f(z) = 0 = f(0)$

Ex. $\dfrac{\operatorname{Img} z^2}{|z|^2}$ at $z=0$

$= \dfrac{2xy}{x^2+y^2} = 2\sin\theta\cos\theta = \sin 2\theta$; at $z=0$; $r=0$ limit doesn't exist.

TS4 Q12

(ii) $\dfrac{Re(z)}{1-|z|}$ at $z=0$

at $f(z)|_{z=0} = 0$.

to show that for each $\varepsilon>0$ $\exists\, \delta>0$ s.t.

$|f(z)| < \varepsilon$ whenever $|z|<\delta$

$$\left|\frac{x}{1-\sqrt{x^2+y^2}}\right| = \left|\frac{r\cos\theta}{1-r}\right|$$

Let $\delta = \dfrac{\varepsilon}{1+\varepsilon}$ $\quad$ If $|z|<\delta$

then $\left|\dfrac{r\cos\theta}{1-r}\right| \le \left|\dfrac{r}{1-r}\right| < \left|\dfrac{\varepsilon/1+\varepsilon}{1-\frac{\varepsilon}{1+\varepsilon}}\right| < \varepsilon$.

Differentiability

Derivatives

Let f be a function whose domain of definition contains a nbd of a point z_0. The derivative of f at z_0, written as $f'(z_0)$ is defined by

$$f'(z_0) = \lim_{z \to z_0} \frac{f(z) - f(z_0)}{z - z_0}$$

provided that the limit exists.

Let define a new complex variable

$$\Delta z = z - z_0$$

$$f'(z_0) = \lim_{\Delta z \to 0} \frac{f(z_0 + \Delta z) - f(z_0)}{\Delta z}$$

Note: Since it is already assumed that f is defined in a neighborhood of z_0, ~~the~~ $f(z_0 + \Delta z)$ is always defined for $|\Delta z|$ sufficiently small.

To make it more simple, let

$$\Delta w = f(z + \Delta z) - f(z)$$

$$f'(z) = \frac{dw}{dz} = \lim_{\Delta z \to 0} \frac{\Delta w}{\Delta z}$$

Example 1. $f(z) = z^2$

$$\lim_{\Delta z \to 0} \frac{\Delta w}{\Delta z} = \frac{(z+\Delta z)^2 - z^2}{\Delta z} = \lim_{\Delta z \to 0} 2z + \Delta z = 2z$$

$$\Rightarrow \frac{dw}{dz} = 2z = f'(z)$$

TS4. Q13. $f(z) = |z|^2$

$$\lim_{\Delta z \to 0} \frac{\Delta w}{\Delta z} = \frac{|z+\Delta z|^2 - |z|^2}{\Delta z} = \frac{(z+\Delta z)(\bar{z}+\Delta\bar{z}) - z\bar{z}}{\Delta z}$$

$$= \frac{z\Delta\bar{z} + \Delta z\bar{z} + \Delta z\Delta\bar{z}}{\Delta z} = \lim_{\Delta z \to 0} \bar{z} + \frac{\Delta\bar{z}}{\Delta z} z + \Delta\bar{z}$$

Let's assume that this limit exists. Then we can approach Δz to 0 in any direction

Let $\Delta z = \Delta x + i0$ direction; $\Delta\bar{z} = \Delta x - i0$

$$\lim_{\Delta z \to 0} \frac{\Delta w}{\Delta z} = \bar{z} + z + \Delta x = \bar{z} + z$$

taking direction $(0, \Delta y)$

$$\lim_{\Delta z \to 0} \frac{\Delta w}{\Delta z} = \bar{z} + (-1) z - i\Delta y = \bar{z} - z$$

Since limit should be unique, now if it exists

$$\bar{z} + z = \bar{z} - z \Rightarrow z = 0.$$

This proves that derivative limit doesn't exist anywhere except 0. To show that it exist at 0

at $z = 0$

$$\lim_{\Delta z \to 0} \frac{\Delta w}{\Delta z} = \lim_{\Delta z \to 0} \Delta\bar{z} = 0$$

Ex. Show that $\bar{z}$ is not differentiable

$$\lim_{\Delta z \to 0} \frac{\overline{z+\Delta z} - \bar{z}}{\Delta z} = \frac{\overline{\Delta z}}{\Delta z} = \frac{\Delta x - i\Delta y}{\Delta x + i\Delta y}$$

If $\Delta x = 0$ then $\lim = -1$; $\Delta y = 0$ then $\lim = 1$.
limit doesn't exist.

TS4 Q1.1
(i) find derivative of $\frac{z-i}{z+i}$ at i

$$\lim_{\Delta z \to 0} \frac{f(z+\Delta z) - f(z)}{\Delta z} = \left.\frac{\frac{z+\Delta z - i}{z+\Delta z+i} - \frac{z-i}{z+i}}{\Delta z}\right|_{z=i}$$

$$= \frac{\Delta z}{\Delta z + 2i} \Big/ \Delta z = \lim_{\Delta z \to 0} \frac{1}{\Delta z + 2i} = \frac{1}{2i}$$

Ex. $(z-2i)^3$ at $5+2i$ $\quad \lim_{\Delta z \to 0} \frac{(5+\Delta z)^3 - 5^3}{\Delta z}$

$$= \lim_{\Delta z \to 0} \Delta z^2 + 15\Delta z + 3 \times 25 = 75$$

Ex $\frac{1.5z+2i}{(3iz-4)}$ at any z $\quad f(z) = \frac{1.5z+2i}{2i(1.5z+2i)} = \frac{1}{2i}$ Constant and hence differentiable

TS4 Q14
(ii) $i(1-z)^n$ at $z=0$

$$\lim_{\Delta z \to 0} \frac{i(1-z-\Delta z)^n - i(1-z)^n}{\Delta z} = \frac{i\left[(1-z-\Delta z)^n - (1-z)^n\right]}{\Delta z}$$

$$= \frac{i\left[(1-z)^n - {}^nC_1 \Delta z (1-z)^{n-1} + {}^nC_2 \Delta z^2 (1-z)^{n-2} \cdots - (1-z)^n\right]}{\Delta z}$$

$$= \lim_{\Delta z \to 0} i\left[-n(1-z)^{n-1} + {}^nC_2 \Delta z (1-z)^{n-2} \cdots\right]$$

$$= -ni(1-z)^{n-1}$$